Final Exam 46921

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Question 0

0 (a) log-likelihood

$$l(\theta) = \sum_{i=1}^{n} \log\left(\frac{1}{\theta x_i^{\frac{1}{\theta}+1}}\right)$$

$$= -\sum_{i=1}^{n} \left(\log\theta + \left(\frac{1}{\theta}+1\right)\log x_i\right)$$

$$\frac{d\,l(\theta)}{d\theta} = -\sum_{i=1}^{n}\left(\frac{1}{\theta} - \frac{\log x_i}{\theta^2}\right)$$

solving $\frac{d\,l(\theta)}{d\theta} = 0$:

$$-\frac{1}{\theta^2}\sum_{i=1}^{n}\left(\theta - \log x_i\right) = 0$$

$$\Rightarrow \hat{\theta} = \frac{1}{n}\sum_{i=1}^{n}\log(x_i)$$

checking 2nd derivative:

$$\frac{d^2 l}{d\theta^2} = -\sum_{i=1}^{n}\left(-\frac{1}{\theta^2}+\frac{2\log X_i}{\theta^3}\right)$$

$$= \frac{1}{\theta^3}\sum_{i=1}^{n}\left(\theta - 2\log X_i\right)$$

$$\overset{\text{plug in }\hat\theta}{=\!=} \frac{1}{\hat\theta^3}\left[\left(\sum_{i=1}^{n}\log X_i\right)-\left(2\sum_{i=1}^{n}\log X_i\right)\right] \le 0$$

so $\hat\theta$ is indeed the MLE.

Q (b) $E(\hat\theta) = \frac{1}{n}\sum_{i=1}^{n} E(\log X_i) = \theta$

$\therefore \hat\theta$ is unbiased,

$$MSE = V(\hat\theta) = \frac{1}{n^2} V\left(\sum_{i=1}^{n}\log X_i\right)$$

$$\overset{\because X_i \text{ iid}}{=\!=} \frac{1}{n^2}\sum_{i=1}^{n} V(\log X_i) = \left(\frac{\theta}{n}\right)^2$$

0 (c) Fisher Info :

$$I(\theta) = E\left(-\frac{\partial^2 \log f(x;\theta)}{\partial\theta^2}\right)$$

$$= E\left(-\frac{1}{\theta^2} + \frac{2\log X}{\theta^3}\right)$$

$$= -\frac{1}{\theta^2} + \frac{2E(\log X)}{\theta^3} = \frac{1}{\theta^2}$$

$\hat{\theta} \sim N\left(\hat{\theta}, \frac{1}{n\hat{\theta}^2}\right)$ by MLE's asym. normality

$\Rightarrow$ CI is $\hat{\theta} \pm Z_{\frac{\alpha}{2}}\left(\frac{1}{\sqrt{n}\,\hat{\theta}}\right)$

Question 1

1 (a) by invariance MLE for $\frac{1}{\lambda} = \overline{X}$

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1(b)

$\hat{\theta} = \bar{X}$, $E(X_i) = \frac{1}{\lambda}$, $V(X_i) = \frac{1}{\lambda^2}$

$\Rightarrow E(\bar{X}) = \frac{1}{\lambda}$, $V(\bar{X}) = \frac{1}{n\lambda^2}$

CLT : $\hat{\theta} \sim N\left(\frac{1}{\lambda}, \frac{1}{n\lambda^2}\right)$

$SE(\hat{\theta}) = \sqrt{V(\bar{X})} = \frac{1}{\sqrt{n}\,\lambda}$

Question 2 : 4

Question 3 : 1

Question 4 : 2

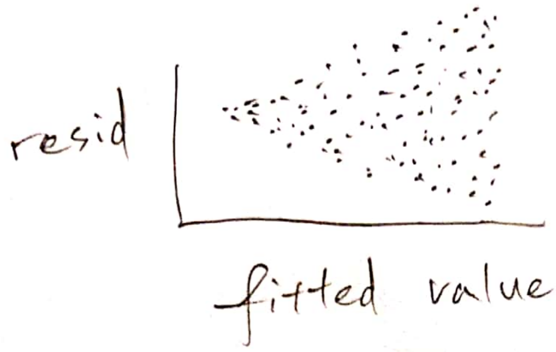
Question 5 : 2

Question 6 : 2

Question 7 : 3

Question 8 : 3

Question 9 :



Question 10 : 1

Question 11 :

These 2 variables look highly correlated.

There is a clear linear relationship

Question 12 :

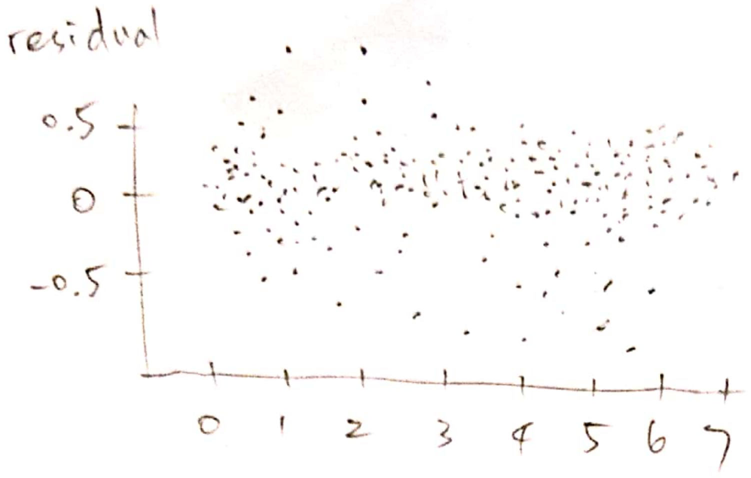
$$CI = \hat{\beta}_1 \pm Z_{0.05} \, SE(\hat{\beta}_1)$$

the reported $\hat{\beta}_1 = 1.0003$. $SE = 0.001$.

$Z_{0.05} = 1.6449$

$\therefore CI = (0.9986551, 1.0019449)$

Question 13



Observed some outliers, among which the linear model overestimates the overnight rates more often than underestimating. These are probably data points from a market condition that has some credit risk concerns and as a result 7-day rates are higher than usual and are not in line with overnight rates.

Considering the sample size is 5000+, the number of outliers is very small. Other than those outliers, the residual v.s. fitted value doesn't show any clear pattern. Overall the simple linear regression seems a good fit.

Question 14 : 3

Question 15

Adding the additional predictor, the AIC becomes smaller (-9637.0089 compared to the original -9635.857).

So the conclusion (simply based on AIC) is the addition of the new predictor is a good idea

Question 16: 2